

Pearson BTEC Higher National qualifications in Computing

**Pearson-set Theme Release
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Unit 16: Computing Research Project

First Teaching from September 2022

First Certification from September 2023

For use with the following qualifications:

Pearson BTEC Level 5

Higher National Diploma in Computing

Pearson BTEC Level 5

Higher National Diploma in Computing for England

(Also for use with Unit 13 – Computing Research Project in HND in Computing, First Teaching Sep 2018)

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1 Introduction to theme

The Pearson-set theme for use with Unit 16: Computing Research Project is:

Artificial Intelligence

Artificial intelligence is at the forefront of innovation within Computer Science that uses a combination of logic, algorithms and large data sets to produce an AI model. The AI model is created to perform specific tasks or make predictions on supplied sets of input data, for example identifying patterns in weather data, internet search data or analysis of medical data. Artificial intelligence is predicted to generate a potential impact to the global economy of \$13 – \$15 Trillion by 2030, with sales of AI related hardware, software and services predicted to see a global revenue of \$900 billion. It is predicted that AI will boost the GDP of China by a little over 26% by 2039, and of North America by 14.5%.

AI requires the input of structured and labelled data, where the outputs are already known. The input data sets to the AI model are intrinsically linked to study field to which the AI engine is to be applied. The AI model can then be used to identify and recognise patterns and relationships within the input data. This identification step is referred to as 'training' the AI model. Once this training is completed, the model can then be used to make predictions and identify patterns within brand new data sets. This new data set can then be added to the existing data set, so that the AI model keeps 'growing'. As the model data set keeps expanding, and the AI algorithms are modified and refined, this gives the impression that the AI is 'learning' and demonstrating 'intelligence'. AI has been used extensively to analyse and process large and complex datasets produced by big data systems, often in real time and using Computer Vision to extract data from image sources.

Developing Artificial Intelligence required a range of knowledge and skills across a broad range of computer science disciplines. AI developers need to be familiar with the algorithms and techniques in fields such as machine learning, natural language processing, computer vision and data science. Knowing the required computing skills will help organisations recruit the correct resources to help develop and extend AI systems.

Artificial Intelligence has a range of benefits across many industry sectors. In the finance industry, AI is rapidly becoming a game changer, using advanced algorithms, models and machine learning to carry out predictive analytics on large, rapidly changing financial datasets to provide more accurate financial predictions. In the field of business operations, AI automation is helping to support and enhance labour

productivity, leading to greater cost savings and increased efficiency. AI is also revolutionising the way businesses interact with their customers, by providing AI driven expert systems to help customers resolve queries as well as providing personalised recommendations based on customer choices and preferences. In the field of biomedical science, AI models help in the development of new drug treatments for a range of diseases by searching and processing large scale medical and DNA datasets.

While Artificial Intelligence has numerous benefits in the analysis and processing of large data sets to solve problems, there are some clear risks to the application technology. AI systems respond to the data fed into the model, and so if this data is not representative of the problem area under study, there is a likelihood that the output of the AI model will be biased. In addition, there are security and privacy concerns on the source and storage of the large datasets used for AI. The rise of the Deepfake image and the manipulation of the human voice is also a concern because of the spread of misinformation. The wide-ranging effects of these risks mean that they can only be dealt with by a diverse range of stakeholders, including computer scientists, law makers, governments, and industry leaders. There are also incidental risks of AI in business, for example the increasing adoption of AI based systems may increase unemployment across a range of sectors and workforce demographics. The theme will enable students to explore some of the topics concerned with Artificial Intelligence from the standpoint of a prospective computing professional or data scientist. It will provide the opportunity for students to investigate the applications, benefits and limitations of Artificial Intelligence while exploring the responsibilities and solutions to the problems it is being used to solve.

2 Choosing a research objective/question

Students are to choose their own research topic for this unit. Strong research projects are those with clear, well focused and defined objectives. A central skill in selecting a research objective is the ability to select a suitable and focused research objective. One of the best ways to do this is to put it in the form of a question. Students should be encouraged by tutors to discuss a variety of topics related to the theme to generate ideas for a good research objective.

The range of topics discussed, could cover the following areas:

- Ethical implications in AI development
- Transparency in AI models
- AI applications
- AI perspectives on governance and regulations

- Bias and fairness in AI algorithms
- AI in cybersecurity; threats and countermeasures

The research objective should allow students to broaden their understanding and widen their perspective of being able to explore, argue, prove, and/or disprove a particular objective. The research objective should be feasible, novel, ethical, relevant and ultimately of interest to the student. Guidance for tutors is available in the ***Pearson-set Assignment Guidance document for Unit 16: Computing Research Project*** and templates are provided for both the research proposal and ethics form.

For those centres who have multiple start dates throughout the academic year, for example students beginning their studies on the Higher National Diploma between January and July, the same theme for both Level 4 and Level 5 Pearson-set units may apply (depending on delivery schedules and when students commence the Pearson-set units). If students are in a position of completing the same theme for both Level 4 and 5, centres must ensure that the theme for the Level 5 is addressed in a different context from the topic selected and applied at Level 4.

Please note that if reasonable adjustments are necessary to meet a specific individual student need you are able to adjust internal assessments to take this into account. Any adjustments must be considered in relation to the centre's policies on equality & diversity and student support.

Further details on how to make adjustments for students with protected characteristics are given in the document '*Pearson Supplementary Guidance for Reasonable Adjustment and Special Consideration in Vocational Internally Assessed Units*' available on our website (<http://qualifications.pearson.com>).

3 Project evidence / outcomes

It is important to recognise that project work is reliant on gathering information/data that can be analysed. The scale of the project means that there must be time for both primary and secondary research. An advised model would be to use secondary research to provide a context for the students to conduct and interpret primary data collection. The project could then yield data that could be compared with the findings of secondary research information.

In assessing the project, the assessor should be able to see a rationale for the project title, an identification of controversial aspects of the title and of the relevant literature/data sources. This will be based primarily on the student's research proposal. Student research should outline the literature/theories that supports the identified research objective/s and include critical evaluation of central arguments paying attention to whether or not the arguments are logically valid. Throughout their research students should be aware of the importance of clear and consistent use of

language and the use of a consistent reference system. Engagement in reflective study of the research process should be evident, with students explaining how their ideas have developed, the significance of results and what they have learnt about the methodology of research. Well edited, focused writing and presentation, where the key decisions, developments, lines of argument and salient research are explained succinctly, is preferable to unstructured writing and presentation where little attempt to select or edit material has been made.

It is important to recognise that there are many different formats that a student could use to present their work and it is important that students think carefully about the suitability of the format in relation to the target audience. Both verbal and written forms of communication should be appropriate to the audience, both in terms of the nature and level of material they use and in terms of length. Students should be guided to produce research that gives a succinct account of the main arguments or developments from their project. If a verbal presentation is the chosen format, the question and answer session should address issues raised by the presentation, but also give students an opportunity to review their work.

Students are to submit as evidence for the unit in addition to their project findings, the **research proposal and ethics form**. The research proposal sets out the plan for how the students will achieve the intended research objective and shows whether the objective will be feasible, ethical and achievable in the time scale. It sets out how secondary research supports the research objective, how the research will be conducted, how the research will be evaluated. Students will need to gain ethical approval before commencing their research, this will be discussed with the tutor during the research proposal.

4 Employer engagement

It is advisable that centres look at the Pearson-set Assignment as an appropriate unit to embed employer engagement, although this is not a mandatory requirement. Developing and establishing links with employers enhances the teaching and learning experience and improves students' employability. Where possible, identifying links with employers as part of the delivery of the Pearson-set Assignment could lead to enhancing and supporting student learning. Real-life projects provide students with the opportunity to develop and acquire appropriate skills, knowledge and expertise required by employers.

5 Sharing of good practice

An appointed External Examiner (EE) for the centre will ask to sample the Pearson-set assignment briefs for review as part of the remote sampling request. Although this is

not a mandatory requirement for centres we strongly advise that centres seek guidance and support from their EE on the Pearson-set assignment. The EE may also include the Pearson-set units in the centre visit sample of student work.

The EE will review and identify exemplars in all aspects of good practice. Good practice will focus on current themes that align to QAA Higher Education Reviews:

- Innovation
- Digital literacy
- Student employability and entrepreneurial skills
- Employer engagement
- Quality of assessment feedback.

6 Assignment guidance / Useful links

Suggested resources and links that centres may find useful are shown below. Centres should choose those resources that are relevant for localised use and complement these with additional resources to support independent research into the chosen topic.

Useful resources for underlying principles, examples of articles and webinars on the theme:

Resource Number	Type of Resource	Resource Titles	Links
1	Article	artificial intelligence (AI)	https://www.techtarget.com/searchenterpriseai/definition/AI-Artificial-Intelligence
2	Article	The Ultimate Guide to Understanding and Using AI Models (2024)	https://viso.ai/deep-learning/ml-ai-models/
3	Blog	What is Labeled Data?	https://www.datacamp.com/blog/what-is-labeled-data
4	Article	A Complete Guide to Data Labeling for AI	https://levity.ai/blog/guide-data-labeling-for-ai
5	Article	What Is Computer Vision? Meaning, Examples, and Applications in 2022	https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-computer-vision/
6	Article	What is Pattern Recognition? A Gentle Introduction (2024)	https://viso.ai/deep-learning/pattern-recognition/

7	Blog	Artificial Intelligence for Big Data Analytics	https://www.vpon.com/en/blogs/artificial-intelligence-for-big-data-analytics/
8	Article	Machine learning, explained	https://mitsloan.mit.edu/ideas-made-to-matter/machine-learning-explained
9	Article	What is natural language processing?	https://www.ibm.com/topics/natural-language-processing
10	Article	A COMPLETE GUIDE TO Natural Language Processing	https://www.deeplearning.ai/resources/natural-language-processing/
11	Article	Artificial Intelligence in Data Science: 5 Definitive Facts	https://hevodata.com/learn/artificial-intelligence-in-data-science/
12	Article	Artificial intelligence: Development, risks and regulation	https://lordslibrary.parliament.uk/artificial-intelligence-development-risks-and-regulation/
13	Blog	The risks of Artificial Intelligence	https://jarnoduursma.com/blog/the-risks-of-artificial-intelligence/
14	Blog	Shedding light on AI bias with real world examples	https://www.ibm.com/blog/shedding-light-on-ai-bias-with-real-world-examples/
15	Report	Increasing Threat of Deepfake Identities	https://www.dhs.gov/sites/default/files/publications/increasing_threats_of_deepfake_identities_0.pdf
16	Journal	The impact of artificial intelligence on human society and bioethics	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7605294/
17	Article	How artificial intelligence is transforming the world	https://www.brookings.edu/articles/how-artificial-intelligence-is-transforming-the-world/
18	Journal	Large AI Models in Health Informatics: Applications, Challenges, and the Future	https://ieeexplore.ieee.org/abstract/document/10261199
19	Journal	Situating methods in the magic of Big Data and AI	https://www.tandfonline.com/doi/abs/10.1080/03637751.2017.1375130
20	Journal	Challenges and opportunities: from big data to knowledge in AI 2.0	https://link.springer.com/article/10.1631/FITEE.1601883

21	Journal	Computer Vision and Image Processing: A Paper Review	https://ijair.id/index.php/ijair/article/view/42
22	Journal	Progress in Neural NLP: Modeling, Learning, and Reasoning	https://www.sciencedirect.com/science/article/pii/S2095809919304928
23	Journal	Taking AI risks seriously: a new assessment model for the AI Act	https://link.springer.com/article/10.1007/s00146-023-01723-z
24	Journal	Confronting the risks of artificial intelligence	https://www.sipotra.it/wp-content/uploads/2019/05/Confronting-the-risks-of-artificial-intelligence.pdf
25	Journal	Governing AI: Understanding the Limits, Possibility, and Risks of AI in an Era of Intelligent Tools and Systems	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3681088
26	Journal	Foundations of AI: The big issues	https://www.sciencedirect.com/science/article/abs/pii/0004370291900480
27	Journal	Formal theories of knowledge in AI and robotics	https://link.springer.com/article/10.1007/BF03037076
28	Book	Principles of Artificial Intelligence	https://link.springer.com/book/9783540113409
29	Report	National AI Strategy	https://assets.publishing.service.gov.uk/media/614db4d1e90e077a2cbdf3c4/National_AI_Strategy_-_PDF_version.pdf
30	Report	Explaining decisions made with AI	https://ico.org.uk/media/about-the-ico/consultations/2616434/explaining-ai-decisions-part-1.pdf
31	Report	Guidelines for secure AI system development	https://www.ncsc.gov.uk/files/Guidelines-for-secure-AI-system-development.pdf
32	Report	GLOBAL OPINIONS AND EXPECTATIONS ABOUT ARTIFICIAL INTELLIGENCE	https://www.ipsos.com/sites/default/files/ct/news/documents/2022-01/Global-opinions-and-expectations-about-AI-2022.pdf
33	Draft AI Act	Artificial intelligence act	https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/698792/EPRS_BRI(2021)698792_EN.pdf

34	Report	Understanding artificial intelligence ethics and safety	https://www.turing.ac.uk/sites/default/files/2019-06/understanding_artificial_intelligence_ethics_and_safety.pdf
35	Presentation	AN INTRODUCTION TO ARTIFICIAL INTELLIGENCE	https://www.uc.edu/content/dam/uc/ce/docs/OLLI/PageContent/ARTIFICIALINTELLIGENCEr.pdf
36	YouTube AI Channel	Two Minute Papers	https://www.youtube.com/@TwoMinutePapers
37	YouTube AI Channel	Sentdex	https://www.youtube.com/@sentdex/featured
38	Webinar	Referencing and writing a bibliography	https://youtu.be/gouEeFKs1xI
39	Webinar	Free Harvard Citation Generator	https://www.citethisforme.com/citation-generator/harvard
40	Webinar	Structuring extended writing	https://youtu.be/o1NMN98GfQc

Additional materials

Pearson-set Report 2018–2019:

<https://hnglobal.highernationals.com/sites/default/files/Pearson-set-Report-2018-2019-final.pdf>

Training Video for the RQF BTEC Higher Nationals Pearson-set Assignments:

https://youtu.be/FkQi_l78_tw

The Pearson-set Assignment Guidance for Unit 16: Computing Research Project should be read in conjunction with the theme release. It provides advice and guidance for both tutors and students.

For any further additional support or queries regarding this document, please email btecdelivery@pearson.com

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